Gluing flag complexes along a tree

Human Compression project

Private draft, 11 October 2026

An abstract simplicial complex is *flag* if each finite set of pairwise adjacent vertices spans a simplex. A subcomplex is *full* if it contains every simplex of the ambient complex whose vertices it contains. We first isolate a standard elementary gluing principle and prove the form needed here. No novelty is claimed for the principle itself.

Lemma 1. *Let T be a tree. For each vertex v let X_v be a flag simplicial complex. For each edge $e = uv$ let X_e be a complex equipped with injective simplicial maps to X_u and X_v, both with full image. Form X by identifying these images. Then each X_v embeds as a full subcomplex of X, and X is flag.*

Proof. An identification of vertices is a finite sequence of identifications across edges of T. Whenever the underlying walk immediately crosses an edge and returns, the two steps cancel, because the edge maps are injective. Reducing the walk proves that a vertex of one X_v cannot be identified with a different vertex of the same X_v. It also proves that the set $S(x)$ of vertices v for which x has a representative in X_v is a connected subtree: representatives along the unique path between any two such vertices are forced and exist.

If two global vertices x, y are represented in both X_u and X_v, they are represented in every block along the path from u to v. If they are adjacent in one of these blocks, fullness of the first edge image puts their edge in that edge complex, and hence in the next block. Repeating this argument transports adjacency along the path. More generally, the identical argument transports every finite simplex. Consequently every block is full in the union.

Now let $x_1, \dots, x_m$ be pairwise adjacent vertices of X. Their support subtrees $S(x_i)$ meet pairwise, since every edge of X comes from a block. Finite families of pairwise intersecting subtrees of a tree have a common vertex. Here is an explicit proof of that fact. Root T at any vertex. In each nonempty subtree take the unique vertex nearest the root, and choose one, p, of greatest distance among these finitely many nearest vertices. For another subtree choose a point where it meets the subtree associated to p. The two nearest vertices lie on the rooted path to that point; the deeper one is p. Connectedness of the other subtree therefore puts p in it. Thus p lies in every subtree.

All x_i are consequently represented in X_p. Adjacency transports to that block as above. Since X_p is flag, these representatives span a simplex in X_p and therefore in X. □

If Y is flag and σ is a simplex, its closed star is a full subcomplex: its vertices are those of σ and all vertices adjacent to every vertex of σ. Any simplex on these vertices joins σ to a clique, and the flag property fills that clique. This observation supplies the full edge inclusions in the next result.

Theorem 2. *Let $\alpha = (A, (A_i)_{i \in I_A})$ and $\beta = (B, (B_i)_{i \in I_B})$ be flag-transitive coset geometries of finite rank. Write $L = I_A \cap I_B$, $P = I_A \setminus L$ and $Q = I_B \setminus L$. Suppose a fixed isomorphism identifies $C = A_P$ with B_Q and identifies $A_{P \cup J}$ with $B_{Q \cup J}$ for every $J \subseteq L$. In $G = A *_C B$ put*

$$G_i = \langle A_{\{i\} \cap I_A}, B_{\{i\} \cap I_B} \rangle \quad (i \in I_A \cup I_B).$$

Then the coset incidence system $(G, (G_i))$ is a flag-transitive geometry. Its chamber stabilizer is the common subgroup $H = A_{I_A} = B_{I_B}$, and for every $J \subseteq I_A \cup I_B$,

$$G_J = A_{J \cap I_A} *_{C \cap A_{J \cap I_A}} B_{J \cap I_B}.$$

Proof. We use the usual tree of the amalgam: vertices are left cosets gA, gB, edges are gC, and gC joins gA to gB. The reduced word theorem proves that this graph is a tree. Using left cosets here only changes the convention by inversion from the usual right-coset geometry.

At an A vertex put the flag complex of α joined to a simplex whose types are Q. At a B vertex put the flag complex of β joined to a simplex of types P. At an edge put the flag complex of the common residue

$$(C, (C \cap A_l)_{l \in L}) = (C, (C \cap B_l)_{l \in L})$$

joined to the simplex of types $P \cup Q$. Flag transitivity identifies this residue with the link of the standard P face in α and with the link of the standard Q face in β. In each endpoint block the edge complex is therefore the closed star of the corresponding face, with the extra simplex already adjoined. These embeddings are full. The construction is transported equivariantly to every coset vertex and edge. Lemma 1 gives a flag complex X.

Every block chamber has one vertex of each type in $I_A \cup I_B$. Flag transitivity in the two input geometries identifies its chambers with the appropriate cosets of H. On an edge, the same identification is by C/H, so these labels agree under every gluing. Thus the chambers of X are naturally G/H: equality of chamber labels is preserved along the unique path of blocks supporting the chamber. In particular G acts transitively on them and their stabilizers are conjugates of H.

We give the same argument for an arbitrary standard face of type J, including the empty face. Its stabilizer in the A block is $D_A = A_{J \cap I_A}$, its stabilizer in the B block is $D_B = B_{J \cap I_B}$, and its stabilizer on the edge is $D_C = C \cap D_A = C \cap D_B$. The equality uses the fixed compatible identification, not merely an abstract isomorphism of these groups. The copies of this face that are identified across edges are exactly the orbit of the standard copy under $\langle D_A, D_B \rangle$. To justify this assertion locally, an edge aC at the A vertex contains the standard J face precisely when its P face aP is incident with the standard $J \cap I_A$ face. Their union is then a flag. Input flag transitivity supplies $d \in D_A$ carrying the standard P face to aP, so $aC = dC$. Thus the supporting incident edges are exactly $D_A C/C$. At the other endpoint they are exactly $D_B C/C$. A path transporting the face crosses only these edge copies, so its consecutive changes of representatives are in D_A or D_B. Conversely each such change fixes the face. The reduced word theorem identifies this generated subgroup with $D_A *_{D_C} D_B$.

Taking a one-vertex face identifies its stabilizer with the stated G_i. Thus vertices of X are precisely the corresponding cosets. Two vertices in a simplex have a common chamber and hence a common coset representative; conversely a common representative places them in a translate of the standard chamber. Their incidence is therefore exactly coset incidence. Taking the whole standard J face gives the displayed intersection formula, because its stabilizer is the intersection of its vertex stabilizers. Finally every simplex of X is contained in a block chamber, so all flags extend to chambers. This proves the geometric assertion. □

Corollary 3. *Under the hypotheses of Theorem 2, the amalgam geometry is residually connected if and only if both input geometries are. It is firm, thin or thick if and only if both inputs have the respective property. Here the group actions need not be faithful.*

Proof. The residue of a standard face of type J has group G_J and parabolics $G_{J \cup \{i\}}$, for $i \notin J$. The intersection formula therefore describes it by the same construction, using the corresponding residues of the two input geometries. If $P \subseteq J$, its A factor is already in C and the amalgam collapses to the B factor. The residue is exactly the residue of β at $J \cap I_B$. If $Q \subseteq J$, the symmetric statement holds. In particular, fixing the standard Q face recovers α, and fixing the standard P face recovers β; an empty such face simply means the whole geometry. This proves necessity of residual connectivity.

For sufficiency, take a residue with at least two types left. The cases just described reduce to residues of the inputs. Otherwise both $P \setminus J$ and $Q \setminus J$ are nonempty. In the tree construction of this residue, each A block is a cone on its nonempty remaining Q simplex, and each B block is a cone on its nonempty remaining P simplex. Each block is connected. The edge complexes contain the nonempty

simplex on $(P \cup Q) \setminus J$, so adjacent blocks meet. Their union over a connected tree is connected. This includes all residues of rank at least two and proves residual connectivity.

For a panel, $J = I \setminus \{i\}$ necessarily contains P or Q. The panel is consequently exactly an input panel by the first paragraph. If $i \in L$, the two possible descriptions agree under the fixed identification in C. Thus the panel sizes are the original indices $[A_{I_A \setminus \{i\}} : H]$ or $[B_{I_B \setminus \{i\}} : H]$, with every original index represented. Comparison with 1, 2 or 3 gives the three assertions. □

Theorem 2 repairs the forward flag-transitivity and parabolic-intersection arguments for compatible parabolic amalgamations in Piedade–Tranchida, arXiv:2505.24662v1, Section 3.2. The geometric hypotheses are essential to this proof: arbitrary special subgroups do not satisfy the corresponding intersection formula, as the separate obstruction note shows. Corollary 3 supplies the resulting residue and panel assertions. The converse flag-transitivity statement for arbitrary input incidence systems and HNN extensions still require their own arguments. This draft has not yet had a distinct mathematical review.